

An Analysis of Speech recognition

Speech recognition

Speech recognition -- Part 1

2000s In the 2000s, DARPA sponsored two speech recognition programs: Effective Affordable Reusable Speech-to-Text (EARS) in 2002, followed by Global Autonomous Language Exploitation (GALE) in 2005. Four teams participated in EARS: IBM; a team led by BBN with LIMS and the University of Pittsburgh; Cambridge University; and a team composed of ICSI, SRI, and the University of Washington. EARS funded the collection of the Switchboard telephone speech corpus, which contained 260 hours of recorded conversations from over 500 speakers. The GALE program focused on Arabic and Mandarin broadcast news. Google's first effort at speech recognition came in 2007 after recruiting Nuance researchers. Its first product, GOOG-411, was a telephone-based directory service. Since at least 2006, the U.S. National Security Agency has employed keyword spotting, allowing analysts to index large volumes of recorded conversations and identify speech containing "interesting" keywords. Other government research programs focused on intelligence applications, such as DARPA's EARS program and IARPA's Babel program. In the early 2000s, speech recognition was dominated by hidden Markov models combined with feed-forward artificial neural networks (ANN). Later, speech recognition was taken over by long short-term memory (LSTM), a recurrent neural network (RNN) published by Sepp Hochreiter & Jürgen Schmidhuber in 1997. LSTM RNNs avoid the vanishing gradient problem and can learn "Very Deep Learning" tasks that require memories of events that happened thousands of discrete time steps earlier, which is important for speech. Around 2007, LSTMs trained with Connectionist Temporal Classification (CTC) began to outperform. In 2015, Google reported a 49 percent error rate reduction in its speech recognition via CTC-trained LSTM. Transformers, a type of neural network based solely on attention, were adopted in computer vision and language modelling, and then to speech recognition. Deep feed-forward (non-recurrent) networks for acoustic modelling were introduced in 2009 by Geoffrey Hinton and his students at the University of Toronto, and by Li Deng and colleagues at Microsoft Research. In contrast to the prior incremental improvements, deep learning decreased error rates by 30%. Both shallow and deep forms (e.g., recurrent nets) of ANNs had been explored since the 1980s. However, these methods never defeated non-uniform internal-handcrafting Gaussian mixture model/hidden Markov model (GMM-HMM) technology. Difficulties analyzed in the 1990s, included gradient diminishing and weak temporal correlation structure. All these difficulties combined with insufficient training data and computing power. Most speech recognition pursued generative modelling approaches until deep learning won the day. Hinton et al. and Deng et al.

Speech recognition -- Part 2

1987 – The back-off model enabled language models to use multiple-length n-grams, and CSELT used HMM to recognize languages (in software and hardware, e.g. RIPAC). At the end of the DARPA program in 1976, the best computer available to researchers was the PDP-10 with 4 MB of RAM. It could take up to 100 minutes to decode 30 seconds of speech. Practical products included:

Speech recognition -- Part 3

$$WRR = 1 - WER = \frac{n - s - d - i}{n} = \frac{h}{n} \quad \{ \displaystyle WRR=1-WER=\frac{(n-s-d-i)}{n}=\frac{h}{n} \}$$

Speech recognition -- Part 4

Neural networks became interesting in the late 1980s before beginning to dominate in the 2010s. Neural networks have been used in many aspects of speech recognition, such as phoneme classification, phoneme classification through multi-objective evolutionary algorithms, isolated word recognition, audiovisual speech recognition, audiovisual speaker recognition, and speaker adaptation. Neural networks make fewer explicit assumptions about feature statistical properties than HMMs. When used to estimate the probabilities of a speech segment, neural networks allow natural and efficient discriminative training. However, in spite of their effectiveness in classifying short-time units such as individual phonemes and isolated words, early neural networks were rarely successful for continuous recognition because of their limited ability to model temporal dependencies. One approach was to use neural networks for feature transformation, or dimensionality reduction. However, more recently, LSTM and related recurrent neural networks (RNNs), Time Delay Neural Networks (TDNN's), and transformers demonstrated improved performance.

Speech recognition -- Part 5

End-to-end learning Since 2014, much research has considered "end-to-end" ASR. Traditional phonetic-based (i.e., all HMM-based model) approaches required separate components and training for pronunciation, acoustic, and language. End-to-end models learn from all the components at once. This simplifies the training and deployment processes. For example, an n-gram language model is required for all HMM-based systems, and a typical 2025-era n-gram language model often takes gigabytes in memory, making them impractical to deploy on mobile devices. Consequently, ASR systems from Google and Apple (as of 2017) deploy on servers and required a network connection to operate. The first attempt at end-to-end ASR was

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the Connectionist Temporal Classification (CTC)-based system introduced by Alex Graves of Google DeepMind and Navdeep Jaitly of the University of Toronto in 2014. The model consisted of RNNs and a CTC layer. Jointly, the RNN-CTC model learns the pronunciation and acoustic model together, however, it is incapable of learning the language model due to conditional independence assumptions, similar to an HMM. Consequently, CTC models can directly learn to map speech acoustics to English characters, but the models make many common spelling mistakes and must rely on a separate language model to finalize transcripts. Later, Baidu expanded on the work with extremely large datasets and demonstrated some commercial success in Mandarin and English. In 2016, the University of Oxford presented LipNet, the first end-to-end sentence-level lipreading model, using spatiotemporal convolutions coupled with an RNN-CTC architecture, surpassing human-level performance in a restricted dataset. A large-scale convolutional-RNN-CTC architecture was presented in 2018 by Google DeepMind, achieving 6 times better performance than human experts. In 2019, Nvidia launched two CNN-CTC ASR models, Jasper and QuartzNet, with an overall performance word error rate (WER) of 3%. Similar to other deep learning applications, transfer learning and domain adaptation are important strategies for reusing and extending the capabilities of deep learning models, particularly due to the small size of available corpora in many languages and/or specific domains. In 2018, researchers at MIT Media Lab announced preliminary work on AlterEgo, a device that uses electrodes to read the neuromuscular signals users make as they subvocalize. They trained a convolutional neural network to translate the electrode signals into words.